

FOSD — Unit 5

Memory Management

GATE CSE PYQ source-page collection • 2017–2026

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- Original question wording, options, tables, code, and figures are preserved on those pages.
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- The selected question numbers are listed below; neighboring questions may remain visible because the original page is preserved.
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Included source pages

CS2020 — source page 6 — selected Q1
CS22024 — source page 17 — selected Q24
CS12024 — source page 35 — selected Q62
CS12025 — source page 14 — selected Q14
CS2-2026 — source page 39 — selected Q55
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Consider the language $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and the following statements.

- I. L is deterministic context-free.
- II. L is context-free but not deterministic context-free.
- III. L is not $LL(k)$ for any k .

Which of the above statements is/are TRUE?

- (A) I only
- (B) II only
- (C) I and III only
- (D) III only

Q.No. 11 Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

- (A) The hole created by first fit is always larger than the hole created by next fit.
- (B) The hole created by worst fit is always larger than the hole created by first fit.
- (C) The hole created by best fit is never larger than the hole created by first fit.
- (D) The hole created by next fit is never larger than the hole created by best fit.

Q.No. 12 Consider the following statements about process state transitions for a system using preemptive scheduling.

- I. A running process can move to ready state.
- II. A ready process can move to running state.
- III. A blocked process can move to running state.
- IV. A blocked process can move to ready state.

Which of the above statements are TRUE?

- (A) I, II, and III only
- (B) II and III only
- (C) I, II, and IV only
- (D) I, II, III, and IV

Q.No. 13

Q.23 Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

- (A) The destination IP address is the IP address of R
- (B) The destination IP address is the IP address of Y
- (C) The destination MAC address is the MAC address of S
- (D) The destination MAC address is the MAC address of Y

Q.24 Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

- (A) Allocate a new page table for a newly created process
- (B) Translate a virtual address to a physical address using the page table
- (C) Raise a trap when a virtual address is not found in the page table
- (D) Raise a trap when a process tries to write to a page marked with read-only permission in the page table

Q.60 The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is _____

Q.61 Consider the following two regular expressions over the alphabet $\{0,1\}$:

$$r = 0^* + 1^*$$

$$s = 01^* + 10^*$$

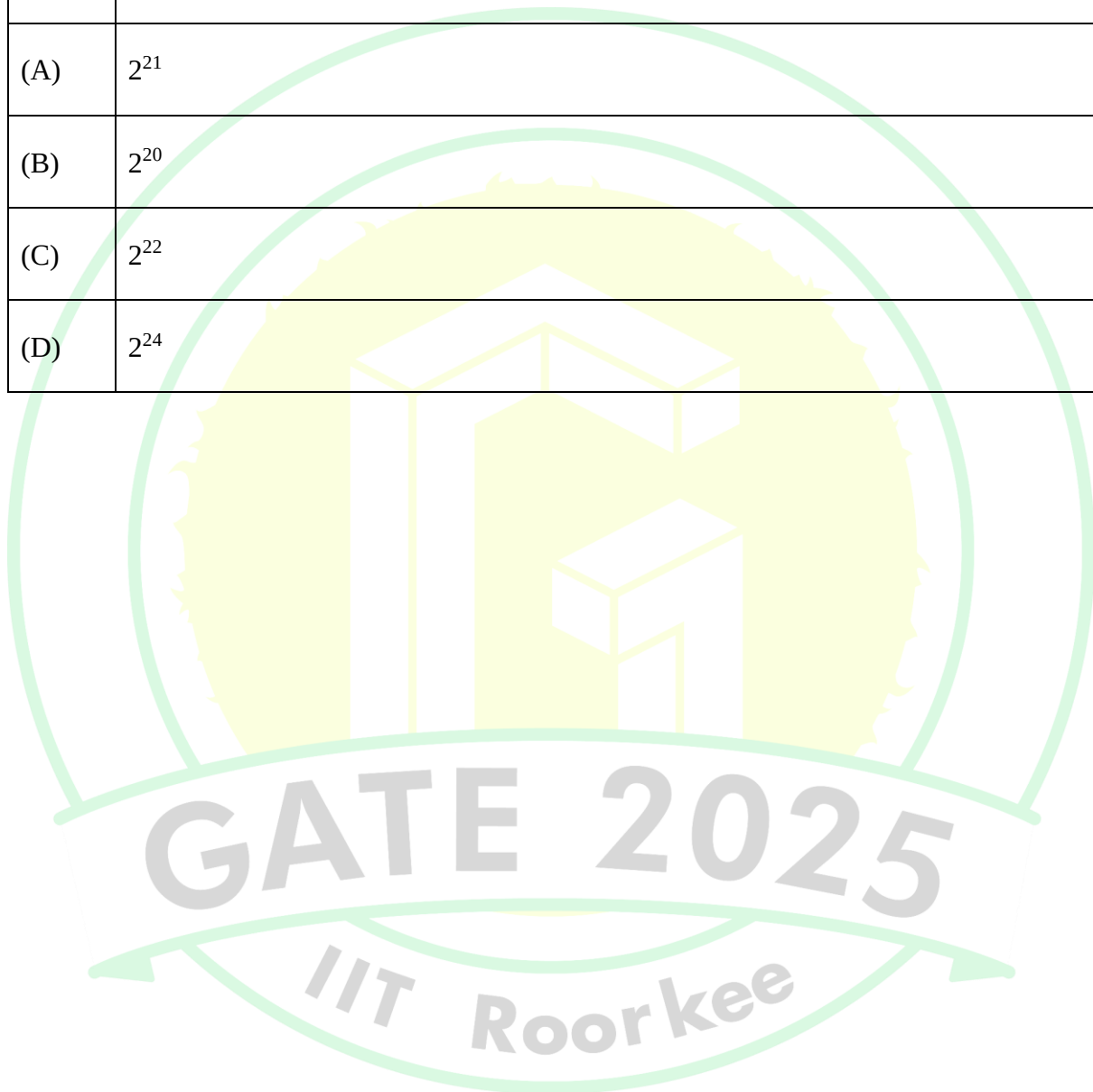
The total number of strings of length less than or equal to 5, which are neither in r nor in s , is _____

Q.62 Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (*in decimal format*) corresponding to the virtual address 2500 (*in decimal format*) is _____

Q.63 A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (*rounded off to 3 decimal places*) that both balls drawn are red is _____



Q.14	Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?
(A)	2^{21}
(B)	2^{20}
(C)	2^{22}
(D)	2^{24}



Q.54	A system has a Translation Lookaside Buffer (TLB) that has a reach of 1 MB. TLB reach is defined as the total amount of physical memory that can be accessed through the TLB entries. The paging system uses pages of size 4 KB. The virtual address space is 64 GB and physical address space is 1 GB. If each TLB entry stores a 4-bit process id, page number, frame number, and a 2-bit control field, then the size of the TLB (in bytes) is _____. (answer in integer) <i>Note:</i> $1K=2^{10}$, $1M=2^{20}$, $1G=2^{30}$
Q.55	Consider contiguous allocation of physical memory to processes using variable partitioning scheme. Suppose there are 8 holes in the memory of sizes 20 KB, 4 KB, 25 KB, 18 KB, 7 KB, 9 KB, 15 KB, and 12 KB. Assume that no two holes are adjacent. Two processes P1 of size 16 KB and P2 of size 9 KB arrive in that order, and they are allocated memory using the best-fit technique. After allocating space to P1 and P2, the number of holes of size less than 8 KB is _____. (answer in integer) <i>Note:</i> $1K=2^{10}$
Q.56	Consider a system with 1 MB physical memory and a word length of 1 byte. The system uses a direct mapped cache, with block numbers starting from 0. The word with physical address $0xA2C28$ is mapped to the cache block number 176_{10}. The maximum possible size of the cache (in KB) for this configuration is _____. (answer in integer) <i>Note:</i> $1K=2^{10}$ and $1M=2^{20}$
Q.57	A non-pipelined instruction execution unit that operates at 1.6 GHz clock takes an average of 5 clock cycles to complete the execution of an instruction. To improve the performance, the system was pipelined with a goal of achieving an average throughput of one instruction per clock cycle. However, it could operate only at 1.2 GHz due to pipeline overheads. While executing a program in the pipelined design, 30% of instructions encountered a stall of 2 cycles due to pipeline hazards. The speed-up obtained by the pipelined design over the non-pipelined one for this program is _____. (rounded off to two decimal places) <i>Note:</i> $1G=10^9$

Q.54	Consider a system that has a cache memory unit and a memory management unit (MMU). The address input to the cache memory is a physical address. The MMU has a translation lookaside buffer (TLB). Assume that when a page is evicted from the main memory, the corresponding blocks in the cache are marked as invalid. For a given memory reference, which of the following sequences of events can NEVER happen?
(A)	TLB miss, Page table hit, Cache hit
(B)	TLB hit, Page table miss, Cache hit
(C)	TLB miss, Page table miss, Cache hit
(D)	TLB miss, Page table miss, Cache miss